Eight channels.
One revolution.

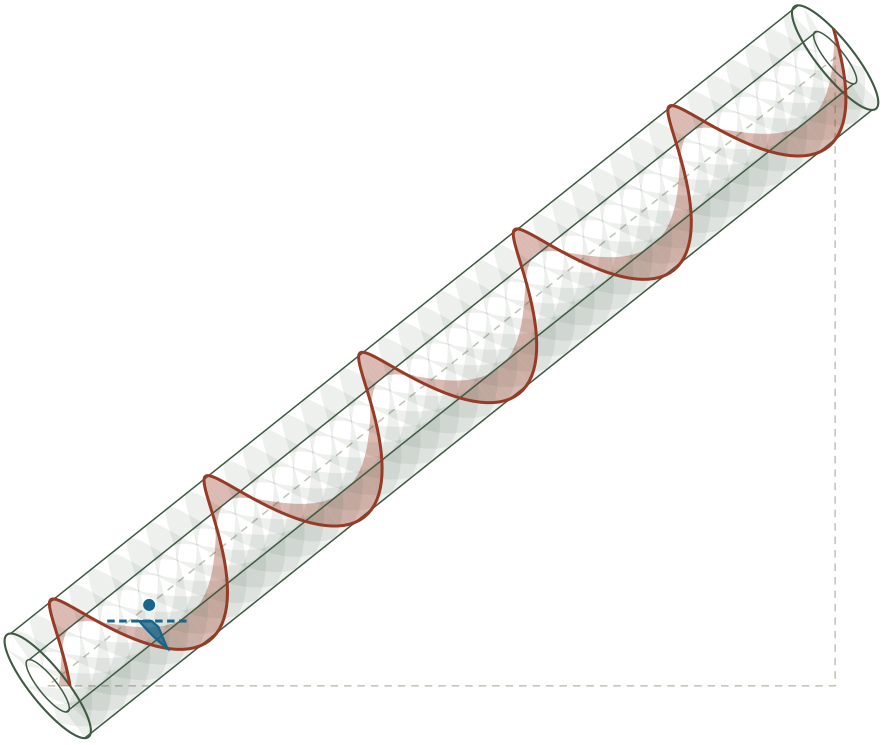
Construct the spiral. Follow one bucket. Read the dimensions.

An ideal, slowly turning, filled screw: watertight rotating casing, zero-thickness blades and no leakage, friction, sloshing or end losses. The blue region is one connected bucket at its spill level. Its volume is a geometrical capacity, not hydraulic efficiency or measured ancient performance. No output rate, human power or safe construction is inferred.

Your retained experiment

8 starts; sixteen-digit source foot; pitch 100% of shaft circumference; angle 36.87°; rotation 0°; cut 50%.

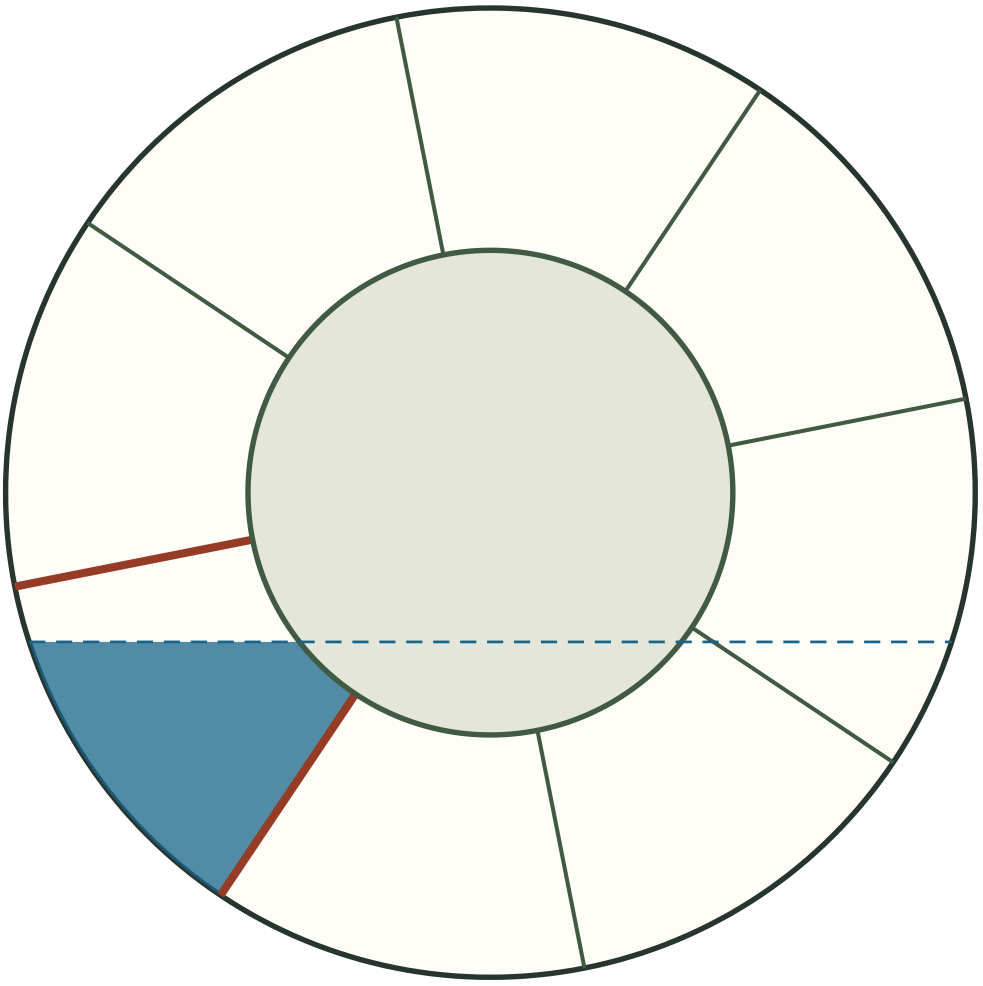
Water fraction 17.0317%; water per turn 1.68096 R^3. R is the outside radius, not a specified ancient metre length. Integration error estimate in fraction 1.15e-10; converged.



Rust: one blade. Blue: bucket cut and horizontal spill level.

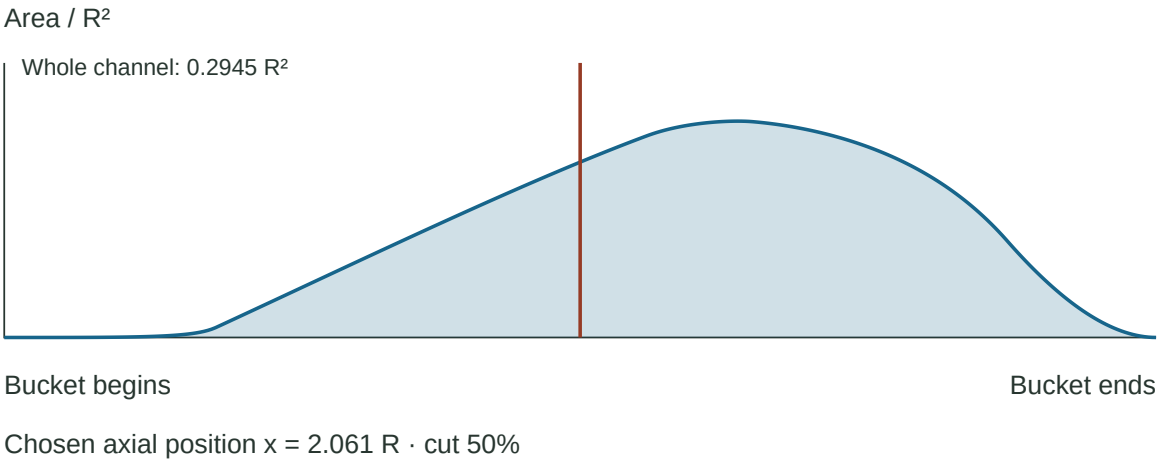
0° turned · bucket advance 0 R · vertical rise 0 R

Cut perpendicular to the shaft



Blue: one bucket only. Other channels’ water is omitted.

Area at this cut: 0.20826 R²

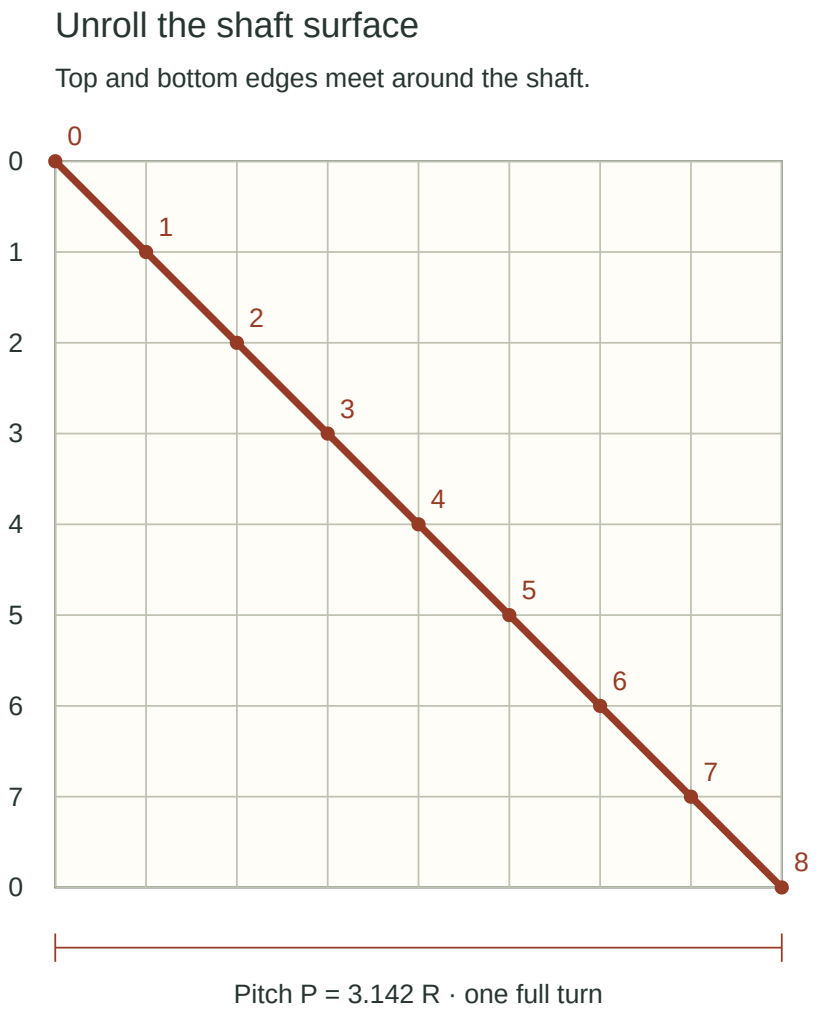


Chosen axial position x = 2.061 R · cut 50%

Same outside radius R and length 16 R; ideal filled geometry

Geometry	Inner / outer radius	Pitch / R	Water fraction	Water / turn (R³)
Source reading	0.5	3.142	17.03%	1.681
Your settings	0.5	3.142	17.03%	1.681
Half your pitch	0.5	1.571	35.76%	1.7647
Quarter your pitch	0.5	0.785	44.33%	1.0937

How one strip becomes eight channels



Grid interval = C/8 = 0.393 R.

Square cells; diagonal 45°.

1. Round the shaft

For every foot of length, give the shaft one digit of diameter. With sixteen digits per foot, shaft diameter = length / 16. The model takes the eventual outer radius as one length unit.

2. Divide its circumference

Quarter, then halve the quarters: eight divisions around each end. These are positions around the same shaft, not eight turns of a single strip.

3. Mark the same interval along it

One eighth of the circumference supplies the axial interval too. Unroll the surface: the source grid has equal sides. Each diagonal step advances both coordinates by one interval.

4. Follow one flexible strip

Advance to the next axial and circumferential crossing. Eight steps bring the strip back to its starting circumferential line, one full circumference farther along the shaft. That axial advance is the pitch P.

5. Repeat; build up the blades

Repeat the helical route from all eight starting divisions. Stack strips until the total diameter reaches length / 8. The ideal drawing gives the blades no thickness; it does not show joinery.

6. Enclose and incline

Boards enclose the channels and rotate with them. The source’s right triangle has rise 3, run 4 and screw length 5. Turn the screw below: one tracked bucket advances one pitch in a full revolution.

Twelve Latin readings

Selected excerpts from Krohn 1912; line-end hyphens joined, paragraph and sentence punctuation normalized; … marks workshop omissions. New English and contextual notes are learning aids. Full pages below preserve the printed text and apparatus.

1. Much water; a modest lift

X.6.1 · p. 237

Est autem etiam cocleae ratio, quae magnam vim haurit aquae, sed non tam alte tollit quam rota.

There is also the screw arrangement, which draws a great quantity of water, but does not raise it as high as the wheel.

cocleae
Genitive of coclea: a snail or spiral, here the water screw. A Greek loan in a Latin text; no Greek original of this passage is supplied.

vim aquae
Vis can mean force or quantity. Here magnam vim aquae is a large amount of water, not a numerical measurement of force.

non tam … quam
‘Not so … as’: tam alte … quam rota compares the height with that of the wheel in the preceding chapter.

Vitruvius gives a comparison, not a measured discharge. The workshop therefore reports volume per revolution in a chosen length unit, not litres per minute.

2. Feet of length, digits of thickness

X.6.1 · p. 237

tignum sumitur, cuius tigni quanta paratur pedum longitudo, tanta digitorum expeditur crassitudo.

Take a timber: however many feet are assigned to its length, assign that many digits to its thickness.

tignum
A timber or beam; after rounding, the inner shaft.

quanta … tanta
A matching pair: ‘as much … so much’. The numerical count is the same, but the units are different.

pedum / digitorum
Genitive plurals of pes and digitus: ‘of feet’ and ‘of digits’. Converting the units supplies the shaft’s slenderness.

We interpret crassitudo here as diameter, following the circular shaft and the conventional reconstruction. No particular metre length is assigned to a Roman foot.

3. Ask the same author what a digit is

III.1.8 · p. 62

relinquitur pes quattuor palmorum. palmus autem habet quattuor digitos. ita efficitur, ut habeat pes sedecim digitos

A foot of four palms remains. A palm has four digits. Thus a foot comes to have sixteen digits.

palmorum
‘Of palms’: four palms make one foot in this account.

sedecim
Sixteen: 4 × 4. This is the conversion used by the source setting.

ita efficitur, ut
‘Thus it comes about that …’; habeat is subjunctive in the resulting clause.

Gwilt’s 1826 English uses ‘inches’ in X.6; Morgan’s 1914 English uses ‘digits’. The twelve-part-foot comparison asks what happens if a modern reader takes ‘inches’ as twelfths. It is our reading experiment, not an alternative stated by Vitruvius.

4. Eight equal parts

X.6.1 · p. 237

id ad circinum rotundatur. in capitibus circino dividentur circumitiones eorum tetrantibus et octantibus in partes octo

Round it with the compass. At the ends, divide their circumferences with the compass, by quarters and eighths, into eight parts.

circinum / circino
Accusative / ablative of circinus: compass; ‘with the compass’ describes the instrument.

tetrantibus et octantibus
By quarters and eighths. Halving a quarter produces an eighth.

capitibus
‘At the ends’: the circular end faces of the shaft, not the ends of eight separate screws.

In the unrolled drawing, the top and bottom edges meet when the sheet wraps around the shaft. The numbered circumferential lines are the same lines at both ends.

5. Make the axial grid equal

X.6.1 · p. 237

et quam magna pars sit octava circinationis tigni, tam magna spatia decidantur in longitudinem. … sic et in rotundatione et in longitudine aequalia spatia fient. … facient decusationes et in decusationibus finita puncta.

Make the intervals along the length as large as an eighth of the timber’s circumference. … Thus the spaces around it and along it will be equal. … They will form crossings, and at those crossings fixed points.

octava
‘An eighth’: it qualifies pars, the circumferential part used as the axial interval.

aequalia spatia
Equal intervals. The unrolled source grid therefore has square cells.

decusationes
Crossings or intersections. They supply points through which the flexible strips are laid.

The grid remains the source’s grid when you alter the pitch. A steeper or shallower diagonal then ceases to follow successive square-grid crossings.

6. A flexible strip, sealed with pitch

X.6.2 · pp. 237–238

sumitur salignea tenuis aut de vitice secta regula, quae uncta liquida pice figitur in primo decusis puncto. deinde traicitur oblique ad insequentis longitudinis et circumitionis decusis

Take a thin willow strip, or one cut from chaste tree, smear it with liquid pitch and fasten it at the first crossing. Then carry it obliquely to the next crossing along the length and around the circumference.

regula
Here a slender wooden strip. Translating every occurrence as a rigid ‘ruler’ would hide how it bends around the shaft.

pice
Ablative of pix: sealing pitch or resin. This Latin word does not name the geometrical pitch P; that is our modern English term.

oblique
Obliquely, across the grid rather than along either family of grid lines.

The excerpt crosses the page break. The whole pages retain Krohn’s apparatus, including variant readings of the crossing instruction.

7. Return to the starting line

X.6.2 · p. 238

et ita pervenit et figitur ad eam lineam recedens a primo in octavum punctum, in qua prima pars est eius fixa.

And so it reaches and is fastened to that line, passing from the first to the eighth point, on which its first part was fixed.

ad eam lineam … in qua
‘To that line … on which’: the return is to the same circumferential generator, farther along the shaft.

a primo in octavum
‘From the first to the eighth’. The counting is compressed; compare the following sentences and the complete page.

fixa
‘Fastened’: feminine because it describes pars, the first part of the strip.

The conventional geometric reading used here takes eight equal diagonal advances to complete a turn: P = shaft circumference. This explicit model interpretation does not settle every textual or construction detail.

8. Eight channels, a snail’s imitation

X.6.2 · p. 238

eadem ratione per omne spatium longitudinis et rotunditatis singulis decusationibus oblique fixae regulae per octo crassitudinis divisiones involutos faciunt canales et iustam cocleae naturalemque imitationem.

In the same way, strips fixed obliquely at the crossings throughout the length and circumference make winding channels through the eight divisions of the thickness: a proper, natural imitation of a snail.

canales
Channels: the spaces bounded by adjacent blades. The model uses eight helical blades to form eight channels.

involutos
Wound or coiled; it describes canales, not eight complete turns in one pitch.

imitationem
An imitation or likeness. The snail is a naming and shape analogy, not a mathematical specification of a biological shell.

One blade advances by P per revolution. Adjacent blade starts are separated axially by P/8. Changing the number of starts below is a modern comparison.

9. Stack up to a total diameter

X.6.3 · p. 238

ita per id vestigium aliae super alias figuntur unctae pice liquida, et exaggerantur ad id, uti longitudinis octava pars fiat summa crassitudo.

Along that trace, further strips are fixed one upon another, smeared with liquid pitch, and built up until the total thickness becomes an eighth of the length.

aliae super alias
‘Others upon others’: further strips follow the same helical trace to build up the blade.

exaggerantur
They are heaped or built up. It does not specify a modern blade’s metal thickness.

summa crassitudo
The total thickness, interpreted here as the finished outside diameter, length / 8.

The source setting consequently has inner radius / outer radius = 1/2. The twelfths experiment changes that ratio to 2/3 while retaining the specified outside diameter.

10. The casing turns with the blades

X.6.3 · p. 238

supra eas circumdantur et figuntur tabulae, quae pertegant eam involutionem. tunc eae tabulae pice saturantur et lamminis ferreis configantur, ut ab aquae vi ne dissolvantur.

Boards are placed and fastened around them to cover the coiling structure. Then the boards are saturated with pitch and secured with iron plates so that the force of the water does not loosen them.

tabulae
Boards: the enclosing outer cylinder. It belongs to the rotating assembly in this reconstruction.

lamminis ferreis
‘With iron plates’: an ablative phrase naming the fastenings.

ne dissolvantur
‘So that they are not loosened’: a negative purpose clause.

The drawing makes this casing transparent so the channels remain visible. It is not a claim that the ancient machine was open on top. This model differs from a screw rotating inside a stationary trough.

11. People turn it by treading

X.6.3 · p. 238

et ita cocleae hominibus calcantibus faciunt versationes.

And in this way the screws make their rotations as people tread.

hominibus calcantibus
An ablative absolute: ‘with people treading’. Calcantibus is a present participle.

versationes
Turnings or rotations. One revolution is the operational unit of our volume calculation.

cocleae
Here nominative plural: the screws are the subject. In the opening passage cocleae was genitive singular.

The sentence does not give a rotation rate, torque or human power. The workshop’s timed animation is a slow teaching motion, not a historical speed estimate.

12. The 3–4–5 right triangle

X.6.4 · p. 238

erectio autem eius ad inclinationem sic erit conlocanda, uti, quemadmodum Pythagoricum trigonum orthogonium describitur, sic id habeat responsum, id est uti dividatur longitudo in partes v, earum trium extollatur caput cocleae; ita erit ab perpendiculo ad imas naris spatium earum partium iiii.

Set its elevation and inclination in the proportions of the Pythagorean right triangle: divide the length into five parts and raise the head of the screw by three of them; the distance from the perpendicular to the lower openings will then be four of those parts.

trigonum orthogonium
Latinized Greek words for a right triangle. Their Greek origin does not make this a surviving Greek passage.

trium / iiii
Three of the five parts rise vertically; four run horizontally. Thus sin α = 3/5 and tan α = 3/4.

naris
Accusative plural here: openings, literally ‘nostrils’. The lower inlet is meant.

At this slope and the source pitch, λ = tan(α)P/(2πR) = 3/8. Modern integration gives about 17.03% of a pitch-length outside cylinder occupied by water; this is not an efficiency percentage.

Model and provenance

Lengths use outside radius R = 1; length L = 16 R. Source shaft diameter L/16 and outside diameter L/8 give inner radius ρ = 1/2. One helical turn advances P = 2πρR at the source setting. At the 3–4–5 slope K = 3/4, λ = KP/(2πR) = 3/8.

At blade phase φ a cut lies at x = P(φ − phase)/(2π). The selected bucket’s level in that cut is w = ρ sin φ_0 − λ(φ − φ_0), with φ_0 = arccos(−λ/ρ) for λ < ρ, otherwise π. Intersect the annular channel with r sin θ ≤ w; retain the connected component around its first bottom. Integrate its area along x, then multiply by the number of starts for water per turn.

Numerical radial/angular boundaries are integrated analytically; an adaptive Simpson integration sums along the shaft. Its error is an estimate, not a rigorous bound. Near the no-retention limit very small values are rounded for display.

Modern analysis: Chris Rorres, The Turn of the Screw: Optimal Design of an Archimedes Screw (2000), DOI 10.1061/(ASCE)0733-9429(2000)126:1(72). This workshop's original diagrams and implementation are not figures or code from the paper.

Original Krohn volume · Gwilt 1826 English · Morgan 1914 English

Reopen this experiment online · Embedded measurements CSV · Embedded 30 mm core wrap

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nario denos aeris constituerunt, et ea re compositio nominis ad hodiernum diem denarium retinet. etiamque quarta pars quod efficiebatur ex duobus assibus et tertio semisse, sestertium vocitaverunt. postea autem quam animadverterunt utrosque numeros esse perfectos, et sex et (10)
decem, utrosque in unum coiecerunt et fecerunt perfectissimum decusis sexis. huius autem rei auctorem invenerunt pedem. e cubito enim cum dempti sunt palmi duo, relinquitur pes quattuor palmorum, palmus autem habet quattuor digitos. ita efficitur, ut habeat pes sedecim digitos (1
et totidem asses aeracius denarius.

9 Ergo si convenit ex articulis hominis numerum inventum esse et ex membris separatis ad universam corporis speciem ratae partis commensus fieri responsum, relinquitur, ut suspiciamus eos, qui etiam aedes deorum inmortalium constituentes ita membra operum ordinaverunt, ut proportionibus et symmetriis separatae atque universae convenientes efficerentur eorum distributiones.

11 Aedium autem principia sunt, e quibus constat figurarum aspectus, ea: primum in antis, quod graece ναος (2
εν παραστασιν dicitur, deinde prostylos, amphiprostylos, peripteros, pseudodipteros, ⟨dipteros⟩, hypaethros. horum **69**
2 exprimuntur formationes his rationibus. in antis erit aedes, cum habebit in fronte antas parietum, qui cellam circumcludunt, et inter antas in medio columnas duas supraque (2
fastigium symmetria ea conlocatum, quae in hoc libro fuerit (5)
perscripta. huius autem exemplar erit ad Tres Fortunas ex
3 tribus quod est proxime portam Collinam. prostylos omnia

---

3 pars quarta *EG* 4 autem *EG*: *om. HS* 7 sexis *HS*: sex eis *EG* auctorem *HG*: ductorem *E*, inventorem *S* 10 sedecim *HS*: XVI EG^c, XIII *G* 11 ases *G* aeracius *HES*, -tius *G* 14 partes *HS* 15 suspiciamus *Perr*: suscipiamus *x* 18 convenientes *rec*: -tesque *x* eorum *H*: operum *EG et in ras. S* 20 ea *Kr*: et HSG^c (*om. G*) naosen parastasin *x* 22 dipteros *add. rec* hypetros *x* 23 *post* rationibus *in x antecipantur verba* quemadmodum et — exemplar (*69, 9 sq.*); *del. Joc*

fluminis inpulsu versatae praestant, quod opus est ad usum.

Eadem ratione etiam versantur hydraletae, in quibus 2 eadem sunt omnia, praeterquam quod in uno capite axis tympanum dentatum est inclusum. id autem ad perpendiculum conlocatum in cultrum versatur cum rota pariter. secundum id tympanum maius ⟨minus⟩ item dentatum planum est conlocatum, quo continetur * * * * ita dentes tympani eius, quod est in axe inclusum, inpellendo dentes tympani plani cogunt fieri molarum circinationem. in qua machina inpendens infundibulum subministrat molis frumentum et eadem versatione subigitur farina.

Est autem etiam cocleae ratio, quae magnam vim haurit VI aquae, sed non tam alte tollit quam rota. eius autem ratio sic expeditur. tignum sumitur, cuius tigni quanta paratur pedum longitudo, tanta digitorum expeditur crassitudo. id ad circinum rotundatur. in capitibus circino dividentur circumitiones eorum tetrantibus et octantibus in partes
258 octo, eaeque lineae ita conlocentur, ut plano posito tigno utriusque capitis ad libellam lineae inter se respondeant, et quam magna pars sit octava circinationis tigni, tam magna spatia decidantur in longitudinem. item tigno plano conlocato lineae ab capite ad alterum caput perducantur ad libellam convenientes. sic et in rotundatione et in longitudine aequalia spatia fient. ita quo loci describuntur lineae, quae sunt in longitudinem spectantes, facient decusationes et in decusationibus finita puncta. his ita emendate descriptis sumitur salignea tenuis aut 2 de vitice secta regula, quae uncta liquida pice figitur in

---

3 hydraletae *Schn*: hydraulae (ydr. *S*) *x* 5 est *Schn*: et (&) *x* 7 *add. Ro* 8 *lacunam sic expl. Joc*: quo continetur axis habens in summo capite subscudem ferream, qua mola continetur 10 plani *Joc*: plane *x* molarum *S*: malarum *HG* 11 infunibulū *x* 15 paratur *Kr*: ratur *HG*, ratus *S* 18 partes *Joc*: pedes *x* 20 utriusque — [crassitudo (*pag. 267, 21*) *om. S* 21 octava *Joc*: -vae *x* 22 dicidantur HG^c, dividantur *G* 25 quo *G*: quod *H* 26 longitudinem *Schn*: -ne *x* 28 aut *G*: ut *H*

primo decusis puncto. deinde traicitur oblique ad insequentis longitudinis et circumitionis decusis, item ex ordine progrediens singula puncta praetereundo et circum involvendo conlocatur in singulis decusationibus, et ita pervenit et figitur ad eam lineam recedens a primo in octavum punctum, in qua prima pars est eius fixa. eo modo quantum progreditur oblique spatium et per octo puncta, tantundem et longitudine procedit ad octavum punctum. eadem ratione per omne spatium longitudinis et rotunditatis singulis decusationibus oblique fixae regulae per octo crassitudinis divisiones involutos faciunt
3 canales et iustam cocleae naturalemque imitationem. ita per id vestigium aliae super alias figuntur unctae pice liquida, et exaggerantur ad id, uti longitudinis octava pars fiat summa crassitudo. supra eas circumdantur et figuntur tabulae, quae pertegant eam involutionem. tunc eae tabulae pice saturantur et lamminis ferreis conligantur, ut ab aquae vi ne dissolvantur. capita tigni ferrea. dextra autem ac sinistra cocleam tigna conlocantur in capitibus utraque parte habentia transversaria confixa. in his foramina ferrea sunt inclusa inque ea inducuntur styli, et ita
4 cocleae hominibus calcantibus faciunt versationes. erectio autem eius ad inclinationem sic erit conlocanda, uti, quemadmodum Pythagoricum trigonum orthogonium describitur, sic id habeat responsum, id est uti dividatur longitudo in partes v, earum trium extollatur caput cocleae; ita erit ab perpendiculo ad imas naris spatium earum partium IIII. qua ratione autem oporteat id esse, in extremo libro eius forma descripta est in ipso tempore.

Quae de materia fiunt organa ad hauriendam aquam, quibus rationibus perficiantur quibusque rebus motus reci-

---

1 insequentis (*Kr*) longitudinis et circumitionis *Mar*: insequentes longitudines et circumitiones *x* 2 decussis *G* 3 singula *ed*: -li *x* puncto *G* 6 in prima qua *G* (*corr. i. m. G*^{c}) 21 ferrea *Ro*: ferret *x* 27 a perpendiculo *Joc*: adppendiculū *x* partium *Mar*: pedes *x* 29 in ipso tempore *x* (*cf. Morgan, Harvard Studies vol. XXI (1910) pag. 18*)

X.6.2–4 · channels, casing and inclination. PDF page 256. SHA-256
7fe0634a712690be5e4dc02739506f6d828d062e0fa784c3268b1b6dfc402a52

pientia praestent versationibus infinitas utilitates, ut essent notiora, quam apertissime potui, perscripta sunt in illo VII tempore. insequitur nunc de Ctesibica machina, quae in altitudinem aquam educit, monstrare. ea fit ex aere. cuius in radicibus modioli fiunt gemelli paulum distantes, habentes fistulas furcillae figura similiter cohaerentes, in medium catinum concurrentes. in quo catino fiunt asses in superioribus naribus fistularum coagmentatione subtili conlocati, qui praeobturantes foramina narium non patiuntur ⟨redire⟩, quod spiritu in catinum est expressum. supra catinum paenula ut infundibulum inversum est 2 attemperata et per fibulam cum catino cuneo traiecto continetur, ne vis inflationis aquae eam cogat elevari. insuper fistula, quae tuba dicitur, coagmentata in altitudine fit erecta. modioli autem habent infra nares inferiores fistularum asses interpositos supra foramina eorum, quae sunt in fundis. ita de supernis in modiolis emboli masculi 3 torno politi et oleo subacti conclusique regulis et vectibus cum molientur qui erit aer ibi cum aqua, assibus obturantibus foramina cogent et trudent inflando pressionibus per fistularum nares aquam in catinum, e quo recipiens paenula spiritu exprimit per fistulam in altitudinem, et ita ex inferiore loco castello conlocato ad saliendum aqua subministratur.

Nec tamen haec sola ratio Ctesibii fertur exquisita, 4 sed etiam plures et variis generibus ab eo liquore pressionibus coactae spiritus efferre ab natura mutuatos effectus ostenduntur, uti merularum [quae motu] voces

---

1 infinitas *Phil*: adinfinitas (*cum* ad *add.*) *x* 4 fit *rec*: sit *x* 5 gemelli G^c: gememelli *x* 6 furcillae sunt figura *x* (sunt *del. Ro*) 7 fiunt *Schn*: fiant *x* 9 praeobdurantes *x* foraminariū *H* 10 *add. Ro* spiritu *Joc*: -tus *x* 11 infudibulū *x* 13 elevari *Schn*: -re *x* 14 altitudinē *G* 19 cum molientur *Kr*: conmoluntur *x* obdurantibus *x* 20 et trudent: extrudent *x* 22 spiritu *Perr*: spiritus *x* 23 inferiore *Joc*: interiore *x* 25 &hesibi *H*, et hę sibi *G* 28 ostenduntur *ed*: -dentur *x* merularum (aquae) *Turn*: merulerūque (quae *G*) *x* *del. Kr* vocis *G*

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